EdgeCount: Statistical Theory and Algorithmic Efficiency

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Abstract

This vignette details the statistical framework underlying the `EdgeCount` package. First, the Random Graph with Given Expected Degrees (RGGED) model is described. Second, the derivation of edge-count statistics is presented. Third, the algebraic optimizations that allow the package to calculate connectedness statistics with linear time complexity are explained. Last, the implementation of vectorized functions contained in the package is discussed, and it is shown that they have optimal time complexity.

Contents

1 Introduction

The `EdgeCount` package provides methods to analyze the connectedness of vertex sets in large, sparse, undirected graphs. Whether analyzing interaction networks or group-element (called term-element) networks, the main question is:

Is an observed number of edges, called edge count, within a set (or between two sets) significantly higher than expected by chance?

To answer this, the package implements a statistical framework built upon the Random Graph with Given Expected Degrees (RGGED) model [5, 6]. The RGGED serves two distinct but related purposes depending on the analysis type:

- Fair comparison (scoring): The RGGED accounts for vertex degrees, which often vary by orders of magnitude. By using analytical expressions for the likelihood of edge counts in this model [15], the package allows for the fair and fast comparison of edge counts across different sets and graph topologies.
- Null model specification:

– For interaction networks (unipartite graphs), the RGGED is utilized directly as the null model. The statistical significance of a set's connectedness is determined by comparing the observed edge count to the distribution under the RGGED on that specific set of vertices.

– For term-element networks (bipartite graphs), the situation is more complex. Because analysis typically involves ranking multiple terms and filtering for existing connections (observed count of at least 1), the RGGED probability acts as a local score or effect size estimator. A second level of modeling—simulating the random selection of element sets—is then required to estimate False Discovery Rates (FDR) and assess global significance.

2 Random graph model

Let $G(V, E)$ be an undirected graph with no loops and no multiple edges.

- Let $N = |V|$ be the order of the graph (number of vertices).
- Let $M = |E|$ be the size of the graph (number of edges).
- Let k_i be the degree of vertex v_i.
- Let $\langle k \rangle$ be the the average vertex degree.

Call the degree sequence of G the nonincreasing sequence of all its vertex degrees. An obvious null model for G is that of a random graph whose realizations always preserve the degree sequence. This model is referred to as the Random Graph with Fixed Degree Sequence (RGFDS). However, it is known that the analytical treatment of the RGFDS is a difficult problem [4, 13, 7, 14]. Namely, deriving simple analytical expressions for the likelihoods of edge-count variables is in general hard with the RGFDS. This implies that slow numerical methods based on simulations must be used [10, 12, 11].

A less constrained random graph model than the RGFDS is the Random Graph with Given Expected Degrees (RGGED) [5, 6]. While an individual realization of the RGGED might not preserve a vertex degree, it is preserved on average over all realizations. The key feature of the RGGED is that edges are treated as independent random variables, enabling the derivation of analytical expressions for the distributions of edge-count variables [15].

In the RGGED model, edges are treated as independent Bernoulli random variables: an edge $v_i v_j$ exists with probability p_{ij} (Bernoulli parameter) and does not exist with probability $1 - p_{ij}$. When the graph is large and sparse on average over its vertices ($\langle k \rangle^2 \ll 2M$), the probability p_{ij} of observing an edge between vertices v_i and v_j is given by [15]:

$$p_{ij} \simeq 1 - \exp\left(-k_i k_j / (2M)\right). \tag{1}$$

Useful upper bounds for p_{ij} are

$$p_{ij} \leq \min\left(1, \frac{k_i k_j}{2M}\right) \leq \frac{k_i k_j}{2M}. \tag{2}$$

2.1 Within and between variables

The `EdgeCount` package considers two types of edge-count variables:

- "within" for the number z of edges observed in a given set S of vertices
- "between" for the number z of edges observed between two sets S_1 and S_2 of vertices

In this framework, the sets and the degrees of their constituent vertices are fixed parameters. Observed values z are thus conditional on these degrees. To an observed value z is associated a corresponding random variable Z in the RGGED, which distribution enables assessment of the significance of the observed value z.

2.2 Edge-count distribution

Edge-count random variables in the RGGED model of a large and sparse graph have approximate Poisson distribution. This is because they are the sum of independent Bernoulli random variables with small parameters $p_{ij} \ll 1$, even when these parameters have different values [9, 8, 15].

Call Z an edge-count random variable defined by the sum of m potential edges, each edge corresponding to a pair $v_i v_j$ of vertices:

$$Z = \sum_{u=1}^{m} B_u \quad \text{with} \quad B_u \sim \text{Bernoulli}\,(p_u = p_{ij})\,. \tag{3}$$

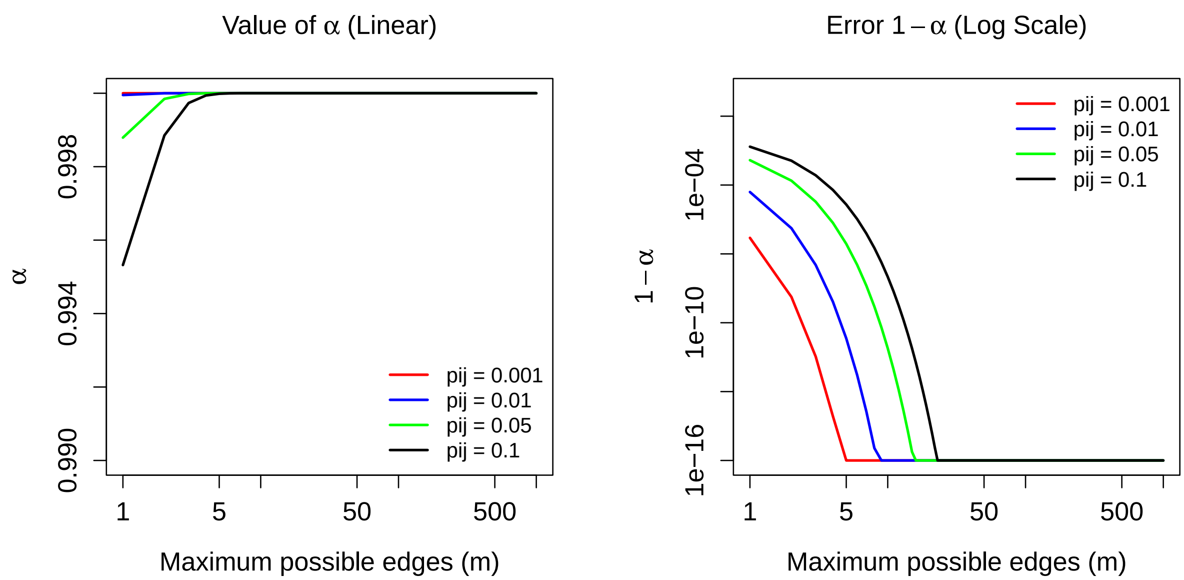
Then, we have $Z \sim \text{Poisson}\,(\lambda, m)$ as defined by

$$\Pr\,(Z = z) = \frac{e^{-\lambda}}{\alpha}\frac{\lambda^z}{z!} \quad \text{with} \quad \lambda = \sum_{u=1}^{m} p_u \quad \text{and} \quad \alpha = e^{-\lambda}\sum_{x=0}^{m}\frac{\lambda^x}{x!}. \tag{4}$$

The normalization factor α is utilized because an observed number z of edges is bounded by a maximum number m of edges. This said, we have $\alpha \simeq 1$ when $p_{ij} \ll 1$. This is numerically illustrated next.

```r
# Function to calculate alpha and its complement
get_alpha_stats <- function(m, pij_avg) {
  lambda <- m * pij_avg
  alpha <- stats::ppois(m, lambda)
  alpha_comp <- stats::ppois(m, lambda, lower.tail = FALSE)
  return(c(alpha, alpha_comp))
}
# Parameters
pij_values <- c(0.001, 0.01, 0.05, 0.1)
m_values <- c(1:29,seq(30, 1000, by = 10))
colors <- c("red", "blue", "green", "black")
# Plots
par(mfrow=c(1, 2), mar=c(5, 5, 4, 2) + 0.1, cex.lab=1.25,
    cex.axis=1.25, cex.main=1.25)
plot(1, type="n", xlab="Maximum possible edges (m)",
     ylab=expression(alpha),
     xlim=c(1, 1000), ylim=c(0.99, 1.00),
     log="x",
     main=expression(paste("Value of ", alpha, " (Linear)")))
for(i in seq_along(pij_values)) {
  lambdas <- m_values * pij_values[i]
  y_vals <- ppois(m_values, lambdas)
  lines(m_values, y_vals, col=colors[i], lwd=2)
}
legend("bottomright", legend=paste("pij =", pij_values),
       col=colors, lwd=2, bty="n", cex=1)
epsilon <- 1e-16
plot(1, type="n", xlab="Maximum possible edges (m)",
     ylab=expression(1 - alpha),
     xlim=c(1, 1000), ylim=c(epsilon, 1),
     log="xy",
     main=expression(paste("Error ", 1 - alpha, " (Log Scale)")))
for(i in seq_along(pij_values)) {
  lambdas <- m_values * pij_values[i]
  y_vals <- ppois(m_values, lambdas, lower.tail = FALSE)
  y_vals <- pmax(y_vals, epsilon)
```

```r
  lines(m_values, y_vals, col=colors[i], lwd=2)
}
legend("topright", legend=paste("pij =", pij_values),
       col=colors, lwd=2, bty="n", cex=1)
```



For a within variable associated with a set S we have

$$m = |S|\left(|S| - 1\right)/2 \quad \text{and} \quad \lambda = \sum_{i<j,\in S} p_{ij}. \tag{5}$$

For a between variable associated with two sets S_1 and S_2 we have

$$m = |S_1|\,|S_2| \quad \text{and} \quad \lambda = \sum_{i\in S_1, j\in S_2} p_{ij}. \tag{6}$$

Since the distribution of an edge-count variable is easily computed, p-values are also calculated.

$$\Pr\left(Z \geq z\right) = \sum_{x=z}^{m} \Pr\left(Z = x\right). \tag{7}$$

P-values are calculated with `R` function `ppois()`.

```r
alpha <- stats::ppois(m, lambda, lower.tail = TRUE)
p_value <- (stats::ppois(z - 1, lambda, lower.tail = FALSE) -
              stats::ppois(m, lambda, lower.tail = FALSE)) / alpha
```

The Poisson distribution enables quantification of the variability on vertex degree modeled by the RGGED. In the model, the degree of a vertex v_i with initial degree k_i is a random variable K_i defined by the edge count between this vertex and all the others:

$$\lambda_i = \frac{k_i}{2M} \sum_{j\neq i} k_j \simeq k_i. \tag{8}$$

That is, K_i has mean k_i and standard deviation $\sqrt{k_i}$. The introduced relative variability on k_i is reasonable: $1/\sqrt{k_i}$.

2.3 Edge-count effect size

As explained earlier, the RGGED is not directly utilized as a null model to answer a question. Rather, it is used as an intermediate step that enables fair and fast comparison of diverse observed edge counts. Such a comparison can be between edge-count variables defined by sets of different sizes. In this case, an effect size is the most appropriate quantity for comparison.

Because edge-count random variables have Poisson distribution, the utilized effect size is

$$\log_2 \text{Anscombe ratio:} \quad lAr\,(z, \lambda) = \frac{1}{2} \log_2 \frac{z + 3/8}{\lambda + 3/8}. \tag{9}$$

The addition of 3/8 stems from the work of Anscombe [2], who showed that $\sqrt{X + 3/8}$ is the optimal variance-stabilizing transformation for a Poisson random variable X.

3 Fast methods for sparse interaction networks

All the code for representation and analysis of undirected unipartite graphs (interaction networks) is contained in S4 clasees `ECGraph` and `ECProb`.

3.1 Sparseness condition

A condition for sparseness in the `EdgeCount` framework is

$$\max_{i,j} \frac{k_i k_j}{2M} < 1. \tag{10}$$

This sparseness condition can be tested with method `summarize_suitability_fast()` of class `ECProb`. The method returns a list containing the proportion `pij_over_1`, among all possible pairs v_i, v_j of vertices, of pairs such that $\text{ve}(k_i k_j)/(2M) \geq 1$. An ideally sparse graph yields `pij_over_1` $= 0$.

Now, approximating p_{ij} values with $k_i k_j/(2M)$ is a conservative approach in the RGGED. Namely, using this approximation

- overestimates p_{ij} values,
- overestimates lambda values,
- overestimates edge-count p-values,
- underestimates effect sizes.

Therefore, utilizing this approximation by default is a reasonable approach. The approximation is called "fast approximation" because it enables faster calculation of edge-count statistics. The speedup is obtained in the computation of lambda parameters.

3.2 Within-set connectedness

Consider a vertex set S of size $|S| = n$. With the fast approximation the λ parameter is

$$\lambda_{in} = \sum_{i<j,\in S} \frac{k_i k_j}{2M}. \tag{11}$$

A naive implementation would have time complexity $\mathcal{O}\left(n^2\right)$. Recall the expression for the square of a sum

$$\left(\sum_{i\in S} k_i\right)^2 = \sum_{i\in S} k_i^2 + 2\sum_{i<j} k_i k_j. \tag{12}$$

Rearranging the cross terms

$$\sum_{i<j} k_i k_j = \frac{1}{2}\left[\left(\sum_{i\in S} k_i\right)^2 - \sum_{i\in S} k_i^2\right], \tag{13}$$

and substituting in the expression for lambda yields

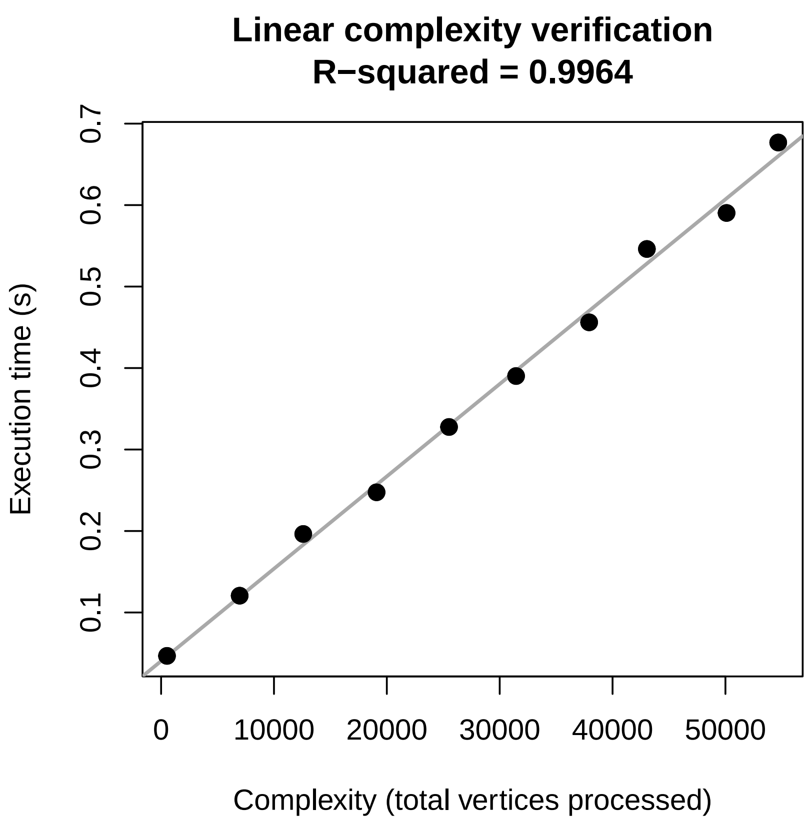
$$\lambda_{in} = \frac{1}{4M}\left[\left(\sum_{i\in S} k_i\right)^2 - \sum_{i\in S} k_i^2\right]. \tag{14}$$

With this expression the computation of λ_{in} has now linear time complexity: $\mathcal{O}(n)$. Method `edge_count_statistics()` of class `ECProb` when used with argument `lambda_method = "fast"` implements this linear computation. Method `calculate_in_stats_fast_vectorized()` is an implementation of "within" edge count statistics for multiple sets provided as input. This method is fully vectorized and uses the `data.table` package [3]. Here is a numerical illustration that the implementation achieves optimal, linear here, time complexity.[1] The illustration is based on the interaction network `sample_ecg` contained in the `EdgeCount` package and derived from the STRING database [16, 1].

```
library(data.table)
library(EdgeCount)
data("sample_ecg")
ecprob <- ECProb(sample_ecg)
n_sim <- 5
n_collections <- 10
collection_size_universe <- seq(10, 1000, length.out = n_collections)
set_size_universe <- 10:100
set.seed(1)

get_collection <- function(target_n_sets){
  n_sets <- round(target_n_sets)
  sets <- lapply(1:n_sets, function(x){
    set_size <- sample(set_size_universe, 1)
    sample(ecprob@names, set_size)
  })
  set_membership_dt <- data.table(
    set_id = rep(seq_along(sets), lengths(sets)),
    element = unlist(sets)
  )
  return(set_membership_dt)
}

collections <- lapply(collection_size_universe, get_collection)
results_list <- lapply(collections, function(set_membership_dt){
  sets_dt <- data.table(set_id = unique(set_membership_dt$set_id))
  times <- numeric(n_sim)
  res <- NULL
  for(k in 1:n_sim){
    start_time <- Sys.time()
    capture.output(
      res <- calculate_in_stats_fast_vectorized(ecprob, sets_dt, set_membership_dt)
    )
    end_time <- Sys.time()
```

[1]Such numerical results are always dependent on the utilized CPU. A faster machine might require larger simulated sets to clearly evidence linearity.

```r
    times[k] <- as.numeric(end_time - start_time, units = "secs")
  }
  complexity <- sum(res$set_size, na.rm = TRUE)
  data.table(complexity = complexity, time = min(times))
})

results_dt <- rbindlist(results_list)

lm_model <- lm(time ~ complexity, data = results_dt)
r_sq <- summary(lm_model)$r.squared

plot(results_dt$complexity, results_dt$time, pch=19, cex=1.2,
     xlab = "Complexity (total vertices processed)",
     ylab = "Execution time (s)",
     main = paste("Linear complexity verification\nR-squared =", round(r_sq, 4)))
abline(lm_model, col = "darkgrey", lwd=2)
points(results_dt$complexity, results_dt$time, pch=19, cex=1.2)
```



The execution time scales linearly (R-squared value close to 1) with the optimal time complexity, which is the sum of the set sizes in each set collection used as input.

The fast approximation of p_{ij} also enables optimal computation of "between" connectedness statistics.

3.3 Between-sets connectedness

Consider two disjoint vertex sets S_1 and S_2 with sizes n_1 and n_2. With the fast approximation, the λ parameter is

$$\lambda_{btw} = \sum_{i \in S_1, j \in S_2} \frac{k_i k_j}{2M}. \tag{15}$$

A naive implementation summing over all pairs would have time complexity $\mathcal{O}(n_1 n_2)$. However, the expression can be factored as the product of two independent sums:

$$\lambda_{btw} = \frac{1}{2M} \left(\sum_{i \in S_1} k_i \right) \left(\sum_{j \in S_2} k_j \right). \tag{16}$$

This factorization reduces the calculation to simply summing the degrees within each set and multiplying the results. Consequently, the computation of λ_{btw} has linear time complexity: $\mathcal{O}(n_1 + n_2)$.

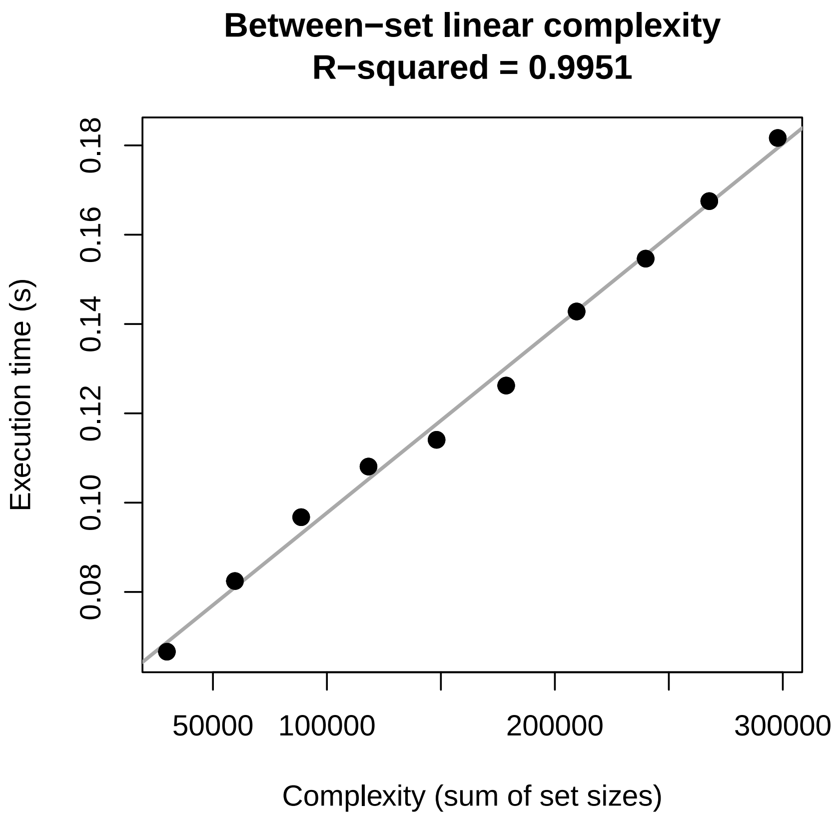
Fast calculation of 'between' edge-count statistics for a single pair of sets is implemented in method `edge_count_statistics()` of class `ECProb` by setting both arguments `set1` and `set2` to non-null vectors and by using `lambda_method = "fast"`.

For multiple pairs of sets, the appropriate method is `calculate_between_stats_fast_vectorized()`. It implements the fast approximation and is fully vectorized in parts by using the `data.table` package [3]. The following simulation confirms that it achieves optimal, linear here, time complexity.[2]

```
library(data.table)
library(EdgeCount)
data("sample_ecg")
ecprob <- ECProb(sample_ecg)

n_sim <- 10
n_collections <- 10
n_pairs_universe <- seq(500, 5000, length.out = n_collections)
set_size_universe <- 10:50
set.seed(123)

get_between_collection <- function(n_pairs){

  n_pairs <- round(n_pairs)
  total_sets <- 2 * n_pairs

  sets <- lapply(1:total_sets, function(x) {
    sample(ecprob@names, sample(set_size_universe, 1))
  })

  set_membership_dt <- data.table(
    set_id = rep(seq_along(sets), lengths(sets)),
    element = unlist(sets)
  )

  pairs_dt <- data.table(
    set1 = seq(1, total_sets, by = 2),
    set2 = seq(2, total_sets, by = 2)
  )
```

[2] Such numerical results are always dependent on the utilized CPU. A faster machine might require larger simulated sets to clearly evidence linearity.

```r
  return(list(pairs = pairs_dt, membership = set_membership_dt))
}

collections_btw <- lapply(n_pairs_universe, get_between_collection)

results_list_btw <- lapply(collections_btw, function(input_data){

  pairs_dt <- input_data$pairs
  membership_dt <- input_data$membership

  times <- numeric(n_sim)
  res <- NULL

  for(k in 1:n_sim){
    start_time <- Sys.time()
    temp_res <- NULL
    capture.output(
      temp_res <- calculate_between_stats_fast_vectorized(ecprob, pairs_dt, membership_dt)
    )
    res <- temp_res
    end_time <- Sys.time()
    times[k] <- as.numeric(end_time - start_time, units = "secs")
  }

  complexity <- sum(res$size1, na.rm = TRUE) + sum(res$size2, na.rm = TRUE)

  data.table(complexity = complexity, time = min(times))
})

results_dt_btw <- rbindlist(results_list_btw)

lm_model_btw <- lm(time ~ complexity, data = results_dt_btw)
r_sq_btw <- summary(lm_model_btw)$r.squared

plot(results_dt_btw$complexity, results_dt_btw$time, pch=19, cex=1.2,
     xlab = "Complexity (sum of set sizes)",
     ylab = "Execution time (s)",
     main = paste("Between-set linear complexity\nR-squared =", round(r_sq_btw, 4)))
abline(lm_model_btw, col = "darkgrey", lwd=2)
points(results_dt_btw$complexity, results_dt_btw$time, pch=19, cex=1.2)
```



4 Fast methods for sparse bipartite networks

Class `ECTermScoring` complements the functionality of `ECGraph` and `ECProb` to the case of undirected bipartite graphs. These represent here membership of elements within groups, called terms in the package.

4.1 Sparseness condition

Call $\mathcal{T}$ the set of vertices that represent terms and $\mathcal{E}$ the set of elements. The sparseness condition is:

$$\max_{i \in \mathcal{T}, j \in \mathcal{E}} \frac{k_i k_j}{2M} < 1. \tag{17}$$

This is the sparseness condition already presented for interaction networks, but adapted to edges existing only between terms and elements. Method `summarize_suitability_fast()` of class `ECTermScoring` is used to test this condition. The method returns a list containing the proportion `pij_over_1`, among all possible pairs v_i, v_j, of values $(k_i k_j)/(2M) \geq 1$. An ideally sparse graph yields `pij_over_1` $= 0$.

But as mentioned for interaction networks, approximating p_{ij} values with $k_i k_j/(2M)$ is a conservative approach in the RGGED. Namely, using this approximation

- overestimates p_{ij} values,
- overestimates lambda values,
- overestimates edge-count p-values,
- underestimates effect sizes.

Therefore, utilizing this approximation by default is a reasonable approach.

4.2 Fast approximation

The fast approximation implemented in `ECTermScoring` is analogous to that used in `ECProb` for between-sets connectedness. Consider a set S of elements and its term neighborhood $\mathcal{T}_S$ (terms connected to at least one element in S).

The expected number of edges λ in this subgraph is given by the double summation:

$$\lambda = \sum_{t \in \mathcal{T}_S} \sum_{e \in S} p_{et}. \tag{18}$$

A naive calculation would have multiplicative time complexity $\mathcal{O}(|S| \cdot |\mathcal{T}_S|)$. However, utilizing the fast approximation $p_{et} = k_e k_t / (2M)$, the expression factors into the product of two sums:

$$\lambda = \frac{1}{2M} \left(\sum_{t \in \mathcal{T}_S} k_t \right) \left(\sum_{e \in S} k_e \right). \tag{19}$$

Consequently, the time complexity for scoring all terms in the neighborhood becomes additive: $\mathcal{O}(|S| + |\mathcal{T}_S|)$. More generally, the average time complexity $T(S)$ for a random set S of elements scales as:

$$T(S) \sim \mathcal{O}\left(|S| \left(1 + \langle k_e \rangle\right)\right), \tag{20}$$

where $\langle k_e \rangle$ is the average element degree across the graph.

4.3 Scoring Terms against Element Sets

The primary analysis on bipartite graphs involves identifying which terms are significantly connected to a given set of elements.

The basic method `terms_ecset_statistics()` calculates connectedness statistics for a single set of element against all its connected terms. These RGGED-based statistics can then be utilized to rank terms for their connectection strength to the set.

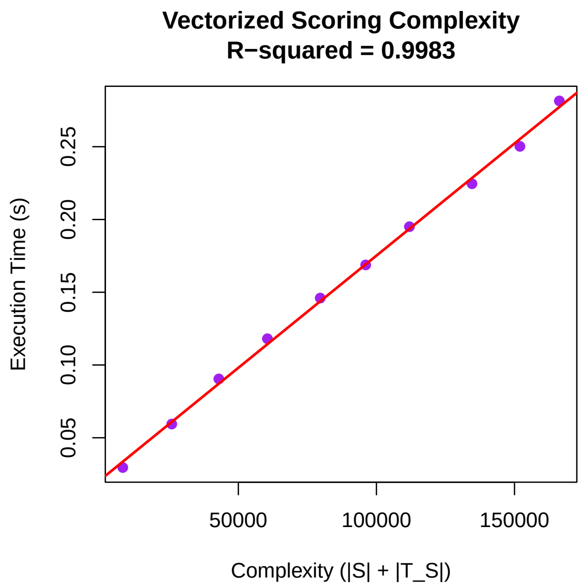
This said, analyses often require scoring multiple sets simultaneously. Method `terms_ecset_statistics_vectorized()` is optimized for this purpose. By utilizing the fast approximation derived in the previous section and `data.table` operations, it achieves linear time complexity. The following simulation verifies this linear scaling.[3] As explained before, linear complexity is the sum of the input elements processed plus the total number of term-element connections evaluated (the neighborhood size).

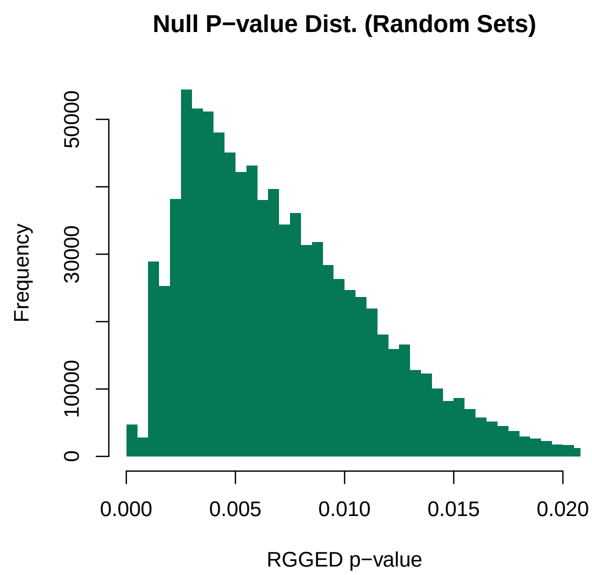
```
library(data.table)
library(EdgeCount)
data("sample_ects")

# --- PART 1: Complexity Simulation ---
n_sim <- 5
n_points <- 10
n_sets_universe <- seq(50, 1000, length.out = n_points)
set.seed(123)

results_list <- lapply(n_sets_universe, function(n_s){
  n_s <- round(n_s)

  # Generate sets with variable sizes (10 to 50)
```

[3] Such numerical results are always dependent on the utilized CPU . A faster machine might require larger simulated sets to clearly evidence linearity.

```r
  current_set_sizes <- sample(10:50, n_s, replace = TRUE)
  random_elements <- unlist(lapply(current_set_sizes, function(sz) {
    sample(sample_ects@elements, sz)
  }))

  input_sets_dt <- data.table(
    set_id = rep(paste0("S", 1:n_s), times = current_set_sizes),
    element = random_elements
  )

  times <- numeric(n_sim)
  res <- NULL

  for(k in 1:n_sim){
    start_time <- Sys.time()
    capture.output(
      res <- terms_ecset_statistics_vectorized(sample_ects, input_sets_dt)
    )
    end_time <- Sys.time()
    times[k] <- as.numeric(end_time - start_time, units = "secs")
  }

  # Exact Complexity: Input size + Output Neighborhood size
  total_neighborhood_size <- sum(vapply(res, nrow, integer(1)))
  true_complexity <- nrow(input_sets_dt) + total_neighborhood_size

  data.table(complexity = true_complexity, time = min(times))
})

results_dt <- rbindlist(results_list)
lm_model <- lm(time ~ complexity, data = results_dt)
r_sq <- summary(lm_model)$r.squared

# --- PART 2: Generate Null Distribution ---
n_hist_sets <- 10000
hist_set_size <- 20
hist_elements <- unlist(replicate(n_hist_sets,
                                  sample(sample_ects@elements, hist_set_size),
                                  simplify = FALSE))
hist_input <- data.table(
  set_id = rep(paste0("H", 1:n_hist_sets), each = hist_set_size),
  element = hist_elements
)
capture.output(
  hist_res <- terms_ecset_statistics_vectorized(sample_ects, hist_input)
)

character(0)

hist_stats <- rbindlist(hist_res)
p_values_sample <- hist_stats$p_value

# --- PART 3: Base R Visualization ---
par(mfrow=c(1, 2), mar=c(5, 4, 4, 2) + 0.1)
```

```r
# Plot A: Linear Complexity
plot(results_dt$complexity, results_dt$time, pch=19, col="purple",
     xlab = "Complexity (|S| + |T_S|)",
     ylab = "Execution Time (s)",
     main = paste("Vectorized Scoring Complexity\nR-squared =", round(r_sq, 4)))
abline(lm_model, col="red", lwd=2)

# Plot B: Non-uniform P-value Distribution
# We use a high number of breaks and limit xlim to see the structure near 0
hist(p_values_sample, breaks = 100, col = "#047857", border = NA,
     main = "Null P-value Dist. (Random Sets)",
     xlab = "RGGED p-value",
     xlim = c(0, 0.02)) # Zoom in to relevant range
```





4.4 Empirical False Discovery Rate

When scoring terms against a set of elements, the package only considers terms that have at least one connection to the set (observed edge count $z \geq 1$). Since unconnected terms ($z = 0$, $p = 1$) are excluded from the output, the distribution of RGGED p-values under the null hypothesis (random sets) is not uniform; specifically, it is depleted near 1.

Consequently, standard multiple testing corrections (like Benjamini-Hochberg) are not applicable. To assess significance rigorously, the package implements an Empirical FDR approach in `terms_ecset_statistics_fdr()`. This method:

1. Generates a null distribution of effect sizes by simulating random element sets of the same size as the input.
2. Calculates a Normalized Enrichment Score (NES) for each term.
3. Estimates the False Discovery Rate (q-value) by comparing the observed NES to the null distribution.

References

[1] Dataset: sample_ecg. `sample_ecg` is a subset of the human STRING network (version 12.0) with interaction scores of 900 or higher. Vertex IDs are Entrez Gene IDs (see Carlson, 2019). The network has been trimmed to create a smaller, more manageable example for the package's vignettes and tests. Source: string-db.org. Data from STRING is licensed under CC BY 4.0., 2025.

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